

Samalka Anandagoda, Ph.D.

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Work authorization: Sponsorship not required.

SUMMARY

- Data scientist and computational researcher with 8+ years of experience analyzing large data sets, building statistical and computational models that resulted in 6+ peer reviewed publications and a software release (SNUGGY).
- Experienced in data preprocessing, feature engineering, hyperparameter tuning, model training and model evaluation for Machine Learning projects such as credit card fraud detection (full list of projects: <https://samalkapiy.github.io/projects/>).
- Ability to effectively communicate technical findings through publications (6+ peer reviewed, full list: <https://samalkapiy.github.io/publications/>), presentations, and cross-disciplinary collaboration.

RELEVANT RESEARCH AND WORK EXPERIENCE

Aug 2024 - Present	Visiting Assistant Professor of Physics -Developed predictive models for real-world ML projects such as Credit card fraud detection which lead to a model accuracy of 99.99% , Space-X model prediction for successful landings which lead to 51 million dollars savings per launch. See full Jupyter notebooks used: https://samalkapiy.github.io/projects/ -Conducted quantitative analysis on ~20 GB of astrophysical data using pandas and NumPy, building computational models, optimizing with scipy to extract meaningful physical insights to answer key questions in research.	Washington College, Maryland
Aug 2023 – Aug 2024	Volunteer Research Fellow Contributed to the development of a GitHub-hosted computational package for simulating multi-source neutrino emissions across the Milky Way Galaxy (Link) utilizing probability distribution functions (PDFs), python programming, and matplotlib.	Clemson University, SC
May 2022 - Aug 2023	Research Assistant Developed and debugged a Monte Carlo particle transport simulation code written in C++. The code incorporates advanced computational algorithms and physics-based simulations to handle complex particle interactions and dynamics. The code was run on a High Performance Computing (HPC) cluster to leverage parallel processing and to increase simulation throughput.	Los Alamos National Laboratory, NM
Aug 2016 - Aug 2023	Astrophysics Researcher (Ph.D. Candidate) Building computational models to answer astrophysical research questions. This project utilized programming skills such as python, parameter fitting using maximum likelihood estimations, writing SQL queries to obtain data from astrophysical catalogs, data visualization and model building.	Clemson University, SC

SOFTWARE RELEASES

SNUGGY: Simulation package that maps the location of the stars in the Milky Way galaxy based on a 2D probability distribution function (PDF) to understand particle interaction processes in the Galaxy. The neutrino and gamma-ray emission are also modeled to ensure the simulated sources follow a log-normal distribution along with a capacity to add more functionality.

SKILLS

Tools	Python (PyTorch, SciKit, pandas, matplotlib, numpy, scipy, seaborn), C++, SQL, Jupyter Notebooks, Google Colab.
Concepts	Statistics, Machine Learning, Clustering algorithms, Regression and Classification, Differential equations, Monte Carlo method. Git version control, High performance computing.

LICENSES AND CERTIFICATIONS

IBM Data Science Professional Certificate

Issued : July 1st 2025, Credential ID: [A7TEPLV81V4K](#)

Skills : data preprocessing, exploratory data analysis (EDA), advanced visualization techniques, and the development of end-to-end machine learning pipelines. Data handling using SQL, pandas. Develop and optimize Decision Tree classifiers; implement K-Nearest Neighbors (KNN) for supervised classification tasks. Train and evaluate Random Forest and XGBoost regression models for performance comparison using cross-validation and metrics such as RMSE, R^2 , and MAE. ([Github link to notebooks](#))

EDUCATION

Aug 2019 - Aug 2023	Ph.D. in Astronomy and Astrophysics	Clemson University, SC
Aug 2016 - Aug 2019	MS in Astronomy and Astrophysics	Clemson University, SC
Nov 2011 - Nov 2015	Bachelor of Science in Physics Major: Physics, Minor: Computer Science and Mathematics	University of Peradeniya, Sri Lanka